

Q1)

B)

- I. It means that the same feature can still be extracted from the image after applying the transformation X where X can be:
 - A. Translation
 - B. Rotation
- II. So that the extracted features can be consistent across different transformations of the same image
- III. **Feature Detector**: Is an algorithm used to extract information by finding local distinctive structures in the image and describe them by surrounding region
Example: Harris Feature Detector
Feature Descriptor: Is a representation of an image that simplifies the image by extracting useful information and throwing away extraneous information.
(lec 4, slide 29)
Example : SIFT

IV.

- Create histogram of local gradient directions at selected scale – 36 bins (each 10 degrees).
- The keypoint is assigned to the orientation of the highest peak.



(lec 5, slide 20)

Then rotate the window to standard orientation

Q2)

B)

- I. Second moment matrix is computed from the image derivatives and is used in the Harris Feature detector to detect the presence of edges or corners
- II. مش علينا
- III. "Bag of Words" is a way to simplify object representation as a collection of their subparts for purposes such as classification.
- IV. مش علينا
- V. مش علينا

Q3) مش علينا

Q4) مش علينا